MILITARY SPECIFICATION

TEST SET, ATOMIC WEAPON: T-4087

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of test set entitled as Test Set, Atomic Weapon: T-4087 which is used with the Honest John warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-108	- Definition of and Basic Requirements for Enclosure of Electric and Electronic Equipment.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

8813397	- Case Combination.
8822072	- Box, Packing, Ammunition, For Test Set, Atomic Weapon: T-4087.
8826477	- Schematic Diagram, Atomic Weapons: T-4087.
8831041	- Bottom Resistor Board Assembly.
8831042	- Chassis Assembly.
8831043	- Selector Switch and Connector Assembly.
8831047	- Test Set Assembly, Atomic Weapons: T-4087.
EL-8842564	- Equipment List.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The test set shall comply with all requirements specified on the drawing (dwg) 8831047, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A six letter serial code shall be assigned to the item. Serial codes required on these items shall be obtained by means of a letter addressed to: Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SARPA-QA-T-M. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial codes required (unused code numbers shall be returned).

3.5 Calibration. The test set shall be capable of self calibration when tested as specified in 4.4.1.

3.6 Self check .

3.6.1 No-go self-check. The test set shall give a red "No-go" indication for all 32 positions of the test selector switch when no-go self-checked. Testing shall be as specified in 4.4.2.1.

3.6.2 Go self-check. The test set shall give a green "Go" indication for all 32 positions of the test selector switch when go self-checked. Testing shall be as specified in 4.4.2.2.

3.7 Operation test limits. The test set shall give the proper Go or No-Go indication when resistances as specified in Table I are connected to the test set. Testing shall be as specified in 4.4.3.

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Table I

Test points for each position of the selector switch		Test resistance extremes for GO and NO-GO indication of the test set for 3 temperature conditions					
Selector Switch Position	Pins on T-4087-2 Cable	Test Resistance Limits (ohms ± 5 percent) -65 ± 10°F GO Test	-65 ± 10°F NO-GO Test	80 ± 20°F GO Test	80 ± 20°F NO-GO Test	160 ± 10°F GO Test	160 ± 10°F NO-GO Test
1	45 to 31	0.35	1.65	0.50	1.95	0.60	2.1
2	11 to 19	2.2	15	2.2	15	2.2	15
3	4 to 3	0.35	1.65	0.50	1.95	0.60	2.1
4	29 to 2	0.35	1.65	0.50	1.95	0.60	2.1
5	28 to 24	0.35	1.65	0.50	1.95	0.60	2.1
6	26 to 17	0.35	1.65	0.50	1.95	0.60	2.1
7	25 to 17	0.35	1.65	0.50	1.95	0.60	2.1
8	36 to 22	0.35	1.65	0.50	1.95	0.60	2.1
9	13 to 12	0.35	1.65	0.50	1.95	0.60	2.1
10	30 to 21	0.35	1.65	0.50	1.95	0.60	2.1
11	23 to 35	0.35	1.65	0.50	1.95	0.60	2.1
12	34 to 21	0.35	1.65	0.50	1.95	0.60	2.1
13	1 to 33	0.35	1.65	0.50	1.95	0.60	2.1
14	46 to 31	0.35	1.65	0.50	1.95	0.60	2.1
15	39 to 40	0.35	1.65	0.50	1.95	0.60	2.1
16	38 to 40	0.35	1.65	0.50	1.95	0.60	2.1
17	6 to 7	0.35	1.65	0.50	1.95	0.60	2.1
18	41 to 2	0.35	1.65	0.50	1.95	0.60	2.1
19	33 to 35	500K	50K	500K	50K	500K	50K
19	22 to 35	500K	50K	500K	50K	500K	50K
20	15 to 16	0.35	1.65	0.50	1.95	0.60	2.10
21	41 to 40	500K	50K	500K	50K	500K	50K
22	15 to 16	500K	50K	500K	50K	500K	50K
22	21 to 16	500K	50K	500K	50K	500K	50K
23	36 to 2	500K	50K	500K	50K	500K	50K
23	15 to 2	500K	50K	500K	50K	500K	50K

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Selector Switch Position	Pins on T-4087-2 Cable	Test Resistance Limits (ohms ± 5 percent)					
		-65 ± 10°F		80 ± 20°F		160 ± 10°F	
		GO Test	NO-GO Test	GO Test	NO-GO Test	GO Test	NO-GO Test
24	2 to 16	40K	10K	40K	10K	40K	10K
25	25 to 2	500K	50K	500K	50K	500K	50K
25	21 to 2	500K	50K	500K	50K	500K	50K
26	3 to 33	500K	50K	500K	50K	500K	50K
26	32 to 33	500K	50K	500K	50K	500K	50K
27	55 to 2	0.35	1.65	0.50	1.95	0.60	2.10
28	33 to 31	500K	50K	500K	50K	500K	50K
28	32 to 31	500K	50K	500K	50K	500K	50K
29	11 to 2	500K	50K	500K	50K	500K	50K
29	10 to 2	500K	50K	500K	50K	500K	50K
30	52 to 2	0.35	1.65	0.50	1.95	0.60	2.10
31	10 to 18	2.2	15	2.2	15	2.2	15
32	10 to 19	2.2	15	2.2	15	2.2	15

Note 1. Test points are indicated on "Schematic Diagram, Atomic Weapons, T-4087", dwg 8826474.
Note 2. Resistance values in ohms: (50K equals 50,000 ohms).
Note 3. Test set indication may be either "GO" or "NO-GO" in the range between the test limits given in table.

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3.8 Continuity test current. The maximum continuity test current shall be less than 100 milliamperes direct current (dc) when tested as specified in 4.4.1.

3.9 Leakage test current. The maximum leakage test current shall be less than 4 milliamperes dc when tested as specified in 4.4.5.

3.10 Selector switch mechanism. The test set shall be capable of making a "NO-GO" indication for a fault in the circuit under test when the test selector switch is switched by hand, without exerting undue force, at the fastest rate possible. Testing shall be as specified in 4.4.6.

3.11 Circuit resistance. The resistance of each continuity circuit of Table II shall be 0.2 ohms maximum when tested as specified in 4.4.7.

Table II

Position of Switches	From Terminal	To Terminal
S2 in 19	5J2-e	5J2-X
S2 in 22	5J2-$\overline{d}$	5J2-p
S2 in 23	5J2-p	5J2-$\overline{a}$
S2 in 25	5J2-$\overline{d}$	5J2-$\overline{h}$
S2 in 26	5J2-$\overline{W}$	5J2-$\overline{FF}$
S2 in 28	5J2-W	5J2-X
S2 in 29	5J2-w	5J2-x
	5J3-$\overline{C}$	Chassis ground

Note 1. Position of switches not listed will not affect the measurement.
Note 2. For connector and pin reference see schematic dwg 8825474.
Note 3. The monitor cable, T-4087-2, shall be connected from 5J2 to the test instrument for tests involving 5J2.

3.12 Insulation resistance. When tested at 500 plus or minus (±) 50 volts direct current (vdc) the insulation resistance of each terminal in Table IIIA to all other terminals in Table IIIA shall be greater than 20 megohms. Testing shall be as specified in 4.4.8.

Table III A

Position of Switches

S1 in "on" position, S2 in position 21, and S3 in "no-go, self-check" position.

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Terminals

5J1-C
5J1-P
5J1-Y
5J1-a
5J1-g
5J1-y
5J1-CC
5J1-FF
5J1 unused terminals, B, D, E, L, M, N, S, T, U, b, c, j, u, v, AA, BB
5J2-B
5J2-C
5J2-F
5J2-G
5J2-H
5J2-J
5J2-P
5J2-R
5J2-V
5J2-W
5J2-z
5J2-AA
5J2-CC
5J2-DD
5J2-EE
5J2-FF
5J2-GG
5J2-X
5J2-Y
5J2-Z
5J2-a
5J2-d
5J2-e
5J2-f
5J2-g
5J2-h
5J2-i
5J2-k
5J2-m
5J2-n
5J2-p
5J2-q
5J2-r
5J2-s
5J2-t
5J2-w
5J2-x
5J2-y
5J2-HH
5J2 unused terminals D, E, L, M, N, S, T, U, b, c, j, u, v, BB
5J3-B
5J3-C

Table III B

Position of Switches	From Terminal	To Terminal
S1 in "on" position	5J2-A	5J3-C
S2 in Position 1	5J2-A	5J1-A
S3 in "operate" position	5J2-J	5J2-K

Note 1: Unused terminals shall be connected together for this test and tested as one terminal.
Note 2: The power switch is off for all insulation tests.
Note 3: Position of switches not listed will not affect the measurement.
Note 4: The monitor cable, T4087-2, shall be connected from 5J2 to the test instrument for tests involving pins of 5J2, and from 5J1 for tests involving pins of 5J1.

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3.13 Operation from 50 Hertz power. The test set shall comply with 3.5 through 3.12 when operated from an input of 115 ± 10 volts root mean square (vrms) and a frequency of 50 Hertz (Hz). Testing shall be as specified in 4.4.9.

3.14 Operation from 400 Hz power. The test set shall comply with 3.5 through 3.12 when operated from an input of 115 ± 10 vrms and a frequency of 400 Hz. Testing shall be as specified in 4.4.10.

3.15 Environmental testing. The test set shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The test set shall also comply with applicable paragraphs in this section before and after each environmental test. Testing shall be as specified in 4.4.11. Tests shall be conducted in the order specified.

3.15.1 Vibration.

3.15.1.1 Cycling. The test set shall be vibrated in its three major axes at the frequencies, temperatures, and gravity force (g) accelerations, specified in Table IV without evidence of damage such as cracks, chips, separations, or loose parts. After vibration the test set shall comply with the requirements of 3.7. Testing shall be as specified in 4.4.11.1.1.

Table IV

Test Cycle Frequency Hz	Cycle Time in Seconds per Cycle	Double Amplitude in Inches	Total Cycling Time - Minutes 80 + 20°F	160 + 5°F	-65 + 5°F
10-33-10*	30 ± 15	±0.06±0.006	15 min**	10 min	10 min
10-55-10*	60 ± 15	±0.03±0.006	15 min	10 min	10 min

*A uniform or logarithmic rate of change of frequency over the test cycle shall be maintained.
**Minimum.

3.15.1.2 Resonance vibration. The test set shall be vibrated for 3 minutes at 80 degrees Fahrenheit (°F) ± 20°F at an amplitude as specified in Table IV and at the most severe resonant node encountered within the frequency range of Table IV when tested as specified in 4.4.11.1.2 without evidence of damage such as cracks, splits, separations, loosening of parts, etc. The time specified in Table IV shall be in addition to the time required for the search for resonance.

3.15.2 Transportation shock. The test set shall withstand shocks of 20 g's along three mutually perpendicular axes without evidence of damage. After shock, the test set shall comply with the requirements of 3.7. Testing shall be as specified in 4.4.11.2.

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3.15.3 High temperature. The test set shall withstand a temperature of 160 + 5°F without cracking, peeling, swelling or other damage and the test set shall comply with 3.7 through 3.10 while at 160 + 20°F when tested as specified in 4.4.11.3.

3.15.4 Low temperature. The test set shall withstand a temperature of -80 + 5°F without cracking, splitting, peeling or other damage and the test set shall comply with 3.7 through 3.10 while at a temperature of -65 + 5°F when tested as specified in 4.4.11.4.

3.15.5 Temperature shock. The test set shall withstand temperature shocks of 160 + 5°F to 80 + 20°F to -65 + 5°F without evidence of cracking, splitting, peeling or other damage, when tested as specified in 4.4.11.5. Upon completion of this test, the test set shall comply with 3.5 through 3.10.

3.15.6 Humidity. The test set shall withstand a relative humidity of 95 + 5 percent at temperatures cycling from 80 + 20°F to 160 + 5°F to 80 + 20°F without cracking, peeling, swelling or other damage when tested as specified in 4.4.11.6. Upon completion of this test, the test set shall comply with 3.7 through 3.10 and 3.12.

3.15.7 Pressure seal, case. The test set case shall show no evidence of leakage when submerged in water. Testing shall be as specified in 4.4.11.7.

3.15.8 Pressure seal, panel. The test set shall not give indication of leakage greater than 2 millimeters of mercury (mm/Hg) in 15 minutes. Testing shall be as specified in 4.4.11.8.

3.16 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.17 Workmanship.

3.17.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.17.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.17.3 Burrs. Parts shall be free from burrs.

3.17.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.17.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.17.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg 8831047) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table V. Tests shall be conducted in order specified in Table V.

Table V

First article inspection and tests

Inspection or Test	Units Tested A	B	C	Non-Dest	Dest
1. Classification of characteristics (see 4.3.2.1)	X	X	X		X
2. Calibration (see 3.5)	X	X	X		X
3. Self-check (see 3.6)	X	X	X		X
4. Operation test limits (see 3.7)	X	X	X		X
5. Continuity test current (see 3.8)	X	X	X		X
6. Leakage test current (see 3.9)	X	X	X		X
7. Selector switch mechanism (see 3.10)	X	X	X		X
8. Circuit resistance (see 3.11)	X	X	X		X
9. Insulation resistance (see 3.12)	X	X	X		X
10. Operation from 50 Hz (see 3.13)	X	X	X		X
11. Operation from 400 Hz (see 3.14)	X	X	X		X
*12. Vibration (see 3.15.1)	X	X	X	X	
*13. Shock (see 3.15.2)	X	X	X	X	
*14. High temperature (see 3.15.3)	X	X	X		X

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*15. Low temperature (see 3.15.4)	X	X	X		X
*16. Temperature shock (see 3.15.5)	X	X		X	
17. Humidity (see 3.15.6)			X	X	
18. Pressure seal, case (see 3.15.7)	X	X	X		X
19. Pressure seal, panel (see 3.15.8)	X	X	X		X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table VI and Table VII), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table VI
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table VII
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspections shall be conducted as follows:

4.3.2.1.1 Selector switch and connector assembly (see dwg 8831043 covering a detail of dwg 8831047).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Leads broken, nicked or without insulation	Visual
102.	Deformation of connector sufficient to impair mating	Visual
103.	Wrong connector	Visual
104.	Void in monitor plug potting greater than 1/8 inch in diameter	Visual/Gage
105.	Gear not pinned to switch shaft	Visual
106.	Any connector socket terminal missing or damaged	Visual
107.	Evidence of poor solder joints (see 3.16)	Visual
108.	Evidence of poor workmanship (see 3.17)	Visual
*109.	Depth of sockets (female sockets of the same diameter must all appear to be within 1/16 inch depth relative to the adjacent sockets)	Visual
110.	Switch will rotate in counter-clockwise direction	Visual/Manual
111.	Any misaligned, loosely assembled, mislocated or missing component	Visual/Manual

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112. Switch binds and will not release.....Visual/Manual
113. Movable switch contacts do not align with fixed switch contacts.....Visual/Manual
114. Objective evidence relating to material missing or improper (see 3.1).....Visual

Minor:
201. Sleeving on wires loose.....Visual
202. Cable binding ties loose.....Visual

*In the event socket inserts do not appear to be within 1/16 inch in depth relative to the adjacent sockets, a gage (see 4.3.4.1) shall be used to determine that variation from the normal depth. Reject any insert that varies by more than 0.063 inch maximum from the normal socket depth of the individual connector. The normal depth of the individual connector sockets shall be determined by calculating the average depth from measurements taken of sockets adjacent to the socket that appears to be defective. The forward edge of the connector shall be utilized as a reference surface for all measurements of socket depths.

4.3.2.1.2 Top resistor board assembly and bottom resistor board assembly prior to potting (see dwgs 8826477 and 8831041 covering a detail of dwg 8831047).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Lack of strain relief on component leads	Visual
	102. Diodes connected to wrong terminals	Visual
	103. Resistors connected to wrong terminals	Visual
	104. Any component missing or incorrect	Visual
	105. Board cracked	Visual
	106. Leads broken or nicked	Visual
	107. Evidence of poor workmanship (see 3.17)	Visual
	108. Evidence of poor soldering (see 3.16)	Visual
	109. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:	None defined.	

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4.3.2.1.3 Top resistor board assembly and bottom resistor board assembly (see dwgs 8826477 and 8831041 covering a detail of dwg 8831047).

Categories		Characteristics	Method of Inspection
Critical:	None defined.		
Major:			
	101.	Potting inadequate	Visual
	102.	Terminals broken off resistor board	Visual
	103.	Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:			
	201.	Marking missing, incorrect or unidentifiable	Visual

4.3.2.1.4 Chassis assembly (see dwg 8831042 covering a detail of dwg 8831047).

Categories		Characteristics	Method of Inspection
Critical:	None defined.		
Major:			
	101.	Leads broken, nicked or without insulation	Visual
	102.	Hardware broken or improperly positioned	Visual
	103.	Evidence of poor workmanship (see 3.17)	Visual
	104.	Evidence of poor soldering (see 3.16)	Visual
	105.	Any component missing or incorrect	Visual
	106.	Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:			
	201.	Marking missing, incorrect or unidentifiable	Visual
	202.	Any misaligned, loosely connected or mislocated component part	Visual/Manual

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4.3.2.1.5 Case, combination (see dwg 8813397 covering a detail of dwg 8831047).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Width, length of panel mounting area open area undersize	Gage
102.	Floating mounting plate nuts missing or broken	Visual
103.	Humidity indicator missing or incorrect	Visual
104.	Rivits missing, cracked or loose	Visual/Manual
105.	Manual relief valve missing or damaged	Visual/Manual
106.	Cover fails to function properly	Visual/Manual
107.	Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:	None defined.	

4.3.2.1.6 Test set assembly atomic weapons T-4087 (see dwg 8831047).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Sealscrew missing or incorrect	Visual
102.	Component(s) missing or incorrect	Visual
103.	Dummy connectors missing incorrect, or incorrectly positioned	Visual
104.	Dust caps missing or incorrect	Visual
105.	Lamp broken or missing	Visual
106.	Dents, scratches of such depth to show parent metal and evidence of damage as to the case	Visual
107.	Evidence of poor workmanship (see 3.17)	Visual
108.	Toggle switch fails to function properly	Visual/Manual
109.	Knobs incorrect or fail to function properly	Visual/Manual
110.	Any misaligned, loosely connected or mislocated parts or assemblies	Visual/Manual

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111. Nest bonded to cover incorrectly or inadequately.....Visual/Manual
112. Objective evidence relating to material missing or improper (see 3.1).....Visual

Minor:
201. Identification plate missing or information contained thereon is incorrect.....Visual
202. Desiccant missing or incorrect.....Visual
203. Serialization missing or improper (see 3.4).....Visual

4.3.2.1.7 Box, packing, ammunition, for test set, atomic weapon: T-4087 (see dwg 8822072).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:
101. Pads missing or incorrectly positioned.....Visual
102. Test set not enclosed within wrapper.....Visual
103. Packing material visibly wet.....Visual
104. Car seal missing, damaged, unsealed or incorrectly positioned.....Visual
105. Metal strapping missing, loose, damaged, or incorrectly positioned.....Visual
106. Rope handles cut or unfastened from box.....Visual
107. Marking on box improper, illegible, smeared, missing or in wrong location.....Visual
108. Evidence of poor workmanship (see 3.17).....Visual
109. Objective evidence relating to material missing or improper (see 3.1).....Visual

Minor:
201. Improper paint color and/or stencil ink color.....Visual
202. Incorrect size of markings.....Visual

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4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Calibration	Major
b. Self-check	Major
c. Operation test limits	Major
d. Continuity test current	Major
e. Leakage test current	Major
f. Selector switch mechanism	Major
g. Circuit resistance	Major
h. Insulation resistance	Major
i. Pressure seal, panel	Major
j. Vibration	Major
k. Shock	Major

Of these tests, j and k are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of tiems required by the contract.

4.3.3.1 Calibration, self check, operation limits, continuity test current, leakage test current, selector switch mechanism, circuit resistance, insulation resistance and pressure seal, panel. Each assembly in the lot shall be tested in accordance with 4.4.1 through 4.4.8 and 4.4.11.8. If any item fails to comply with 3.5 through 3.12 and 3.15.8, the item shall be rejected and removed from the lot.

4.3.3.2 Vibration and shock. A sample shall be selected at random in accordance with Table VIII. The sample shall be tested in accordance with 4.4.11.1 and 4.4.11.2. If any item fails to comply with 3.15.1 and 3.15.2, the lot shall be rejected.

Table VIII

Sample sizes for destructive testing

Number Required to be Delivered	Sample Size for No Failures
2-8	1
9-180	2
181-500	3

Lot Size Required = Number Required to be Delivered + Sample Size.

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4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8842564). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Calibration. Connect the test set to the power source. Place the test selector switch in position 1 and the calibrate knob in the start position. With monitor cable, T-4087-2, connected from the monitor connector to the self-check connector, and the self-check switch in the GO self-check position, turn the power switch on. Allow 5 minutes warm-up. The red NO-GO lamp should be on. Slowly turn the calibrate knob clockwise from the start position until the greed GO lamp lights. Turn the self-check switch to the NO-GO self-check position and the red lamp should light. Re-check by repeated switching between GO and NO-GO self-check positions for positive operation of the green and red lights. If improper lamp indication occurs, turn the calibrate knob slightly clockwise to obtain a GO indication and slightly counterclockwise to obtain a NO-GO indication. When the test set gives the proper indication for each position of the self-check switch, lock the calibrate knob.

4.4.2 Self-check.

4.4.2.1 NO-GO self-check. After calibration in accordance with 4.4.1 turn the self-check switch to the NO-GO position (self-check). With the monitor cable connected from the monitor connector to the self-check connector, turn the test selector switch through all 32 positions stopping on position 32. The red NO-GO lamp must light at each position showing that the test set is capable of revealing a fault if one exists in any unit under test.

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4.4.2.2 GO self-check. After calibration and NO-GO self-check in accordance with 4.4.1 and 4.4.2.1 turn the self-check switch to GO self-check. With the test selector switch in position 32, the green GO lamp should light. Turn the test selector switch through all 32 positions again. This time the green GO lamp must light in each position showing that the test set has no internal fault which would give a NO-GO indication for a satisfactory unit under test.

4.4.3 Test resistance limits. Connect the test set to a 107 + 2 vrms 60 Hz, power source with power cable T-4087-2. Calibrate and self-check the test set in accordance with 4.4.1, 4.4.2.1 and 4.4.2.2. Use the monitor cable T-4087-2 to connect the monitor connector on the test set to a test circuit capable of placing either the GO or NO-GO resistance between the test points of Table I for all 32 test selector switch positions. Turn the self-check switch to the operate position and the test selector switch to the number 1 or start position. Connect the NO-GO resistance value to the test points of the number 1 selector switch position as listed in Table I. The red NO-GO lamp must light. Switch the GO resistance value from Table I in place of the NO-GO resistance. This must be done without shorting or opening the test points. The green GO lamp must light. Change the input voltage to 123 + 2 vrms, 60 Hz. The green GO lamp must remain on. Switch back to the NO-GO resistance value of Table I. The red NO-GO lamp must light. Repeat this procedure except for calibration and self-check for all 32 positions of the test selector switch.

4.4.4 Continuity test current. Calibrate and self-check the test set in accordance with 4.4.1, 4.4.2 and 4.4.2.2. With one end of the monitor cable connected to the test set monitor connector, connect a suitable dc current measuring instrument between pins 11 and 19 (19 positive) of the free end of the monitor cable. The resistance of the meter leads and meter itself must not total more than 0.15 ohms. Turn the self-check switch to "operate" and the test selector switch to position number 2. With an input power of 123 + 2 vrms, 60 Hz, the current as measured by the meter must be less than 100 milliamperes dc. A short jumper shall be placed from pin 50 to 51 of the monitor cable.

4.4.5 Leakage test current. Connect one end of the monitor cable, T-4087-2, to the test set monitor connector. Connect a suitable dc current measuring instrument with an input impedance of less than 150 ohms between pins 40 and 41 (41 negative) of the free end of the monitor cable, T-4087-2. Connect the test set to a 123 + 2 vrms, 60 Hz, power source. Turn the test selector switch to position 21. The current as measured by the meter must be less than 4 milliamperes dc.

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4.4.6 Selector switch mechanism. Calibrate the test set in accordance with 4.4.1 and self-check in accordance with 4.4.2.1 and 4.4.2.2. Connect the monitor cable, T-4087-2, from the monitor connector on the test set to a test circuit which will cause the test set to give a NO-GO indication in positions 7, 16, 22, and 31 and a GO indication in the remaining positions. Turn the test selector switch by hand without exerting undue force, at the fastest rate possible. A GO indication must be given at every position except 7, 16, 22, and 31. A NO-GO indication must be given each time one of the positions with a fault is passed. The indication will be only momentary unless the switch stops on a fault position. The self-check position must be in the "operate" position during this test.

4.4.7 Circuit resistance. The resistance between the terminals specified in Table II shall be determined with a suitable electrical resistance measuring instrument accurate to 0.001 ohms.

4.4.8 Insulation resistance. The insulation resistance between terminals specified in Table III A and Table III B shall be determined with an insulation tester, capable of supplying the required voltage and having a scale reading directly in megohms. The current shall be limited to a maximum of 5 milliamperes to protect human life.

4.4.9 Fifty Hz operation. Repeat the procedures of 4.4.3 except use a 50 ± 2 Hz power source in place of the 60 ± 2 Hz power source.

4.4.10 Four-hundred Hz operation. Repeat the procedures of 4.4.3 except use a 400 ± 4 Hz power source in place of the 60 + 2 Hz power source.

4.4.11 Environmental testing (see 3.15).

4.4.11.1 Vibration. The test set shall be fastened securely on a suitable vibration producing machine such as to permit vibration in the direction of the three major axes. The test set should be shielded from strong magnetic fields during vibration. Vibration tests shall be conducted under both resonant and cycling conditions as directed in 4.4.11.1.1 and 4.4.11.1.2. At the end of the test period the test set shall be inspected thoroughly for damage or defects resulting from the vibration test.

4.4.11.1.1 Cycling. The test set shall be vibrated in the directions of three major axes at the frequencies, amplitudes, and for the times specified in Table IV. Following this test, the test set shall be tested as specified in 4.4.3.

4.4.11.1.2 Resonance vibration. The test set shall be removed from its case to facilitate determination of resonance. Resonant frequencies of the test set shall be determined in each of the three major and mutually perpendicular axes by varying the frequency of applied vibration slowly through the 10 to 55 Hz frequency range at double amplitudes

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not exceeding those given in Table IV. The test set shall be vibrated in each of the three major and mutually perpendicular axes at 80 + 20°F as specified in Table IV for 3 minutes at the most severe resonant node encountered. If no resonance is found, the test set shall be vibrated for a total of 9 minutes (3 minutes in each axis) at a frequency of 55 Hz and an amplitude of + 0.030 + 0.006 inches.

4.4.11.2 Transportation shock. The test set shall be installed in its case and each of the six sides of the case shall then be successively mounted to the shock testing machine. In each mounted position on the shock tester, three shocks shall be applied. These shocks shall be 20 + 2 g shocks of 11 + 1 milliseconds half amplitude duration. The test set shall not be energized during the test. Following this test, the test set shall be tested as specified in 4.4.3.

4.4.11.3 High temperature. The test set shall be placed within a chamber and the internal temperature of the chamber gradually raised to 160 + 5°F. The chamber shall have an internal relative humidity of not more than 5 percent. The test set shall be maintained at 160 + 5°F for a minimum period of 2 hours. At the conclusion of this exposure period and while still at this temperature, the test set shall be tested as specified in 4.4.3 through 4.4.6.

4.4.11.4 Low temperature. The test set shall be placed within a chamber, and the chamber gradually cooled to and maintained at a temperature of -80 + 5°F for a minimum period of 4 hours. The temperature of the chamber shall then be raised to -65 + 5°F and stabilized (see 6.4) or maintained for 2 hours whichever is longer. At the conclusion of this exposure period and while still at this temperature, the test set shall be tested as specified in 4.4.3 through 4.4.6.

4.4.11.5 Temperature shock. The test set shall be placed within a test chamber having a temperature of 160 + 5°F. The test set shall be subjected to this temperature for a period of 4 hours, at the conclusion of which, and within 5 minutes, the item shall be transferred to a chamber having an internal temperature of 80 + 20°F. The item shall be subjected to this temperature for a period of 4 hours, after which and within 5 minutes, the item shall be transferred to a chamber having an internal temperature of -65 + 5°F. The item shall be subjected to this temperature for a period of 4 hours. At the conclusion of this period and within 5 minutes, the item shall be transferred to the 80 + 20°F test chamber and the item shall be subjected to this temperature for a minimum of 4 hours. This completes one cycle. The number of complete cycles shall be three. Following the third cycle, the test set shall be tested as specified in 4.4.1 through 4.4.6.

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4.4.11.6 Humidity. The test set, with its case in the closed condition, shall be placed in a test chamber having a temperature of 80 + 20°F. During the first 2 hour period, the temperature shall be gradually raised to 160 + 5°F. This temperature shall be maintained during the next 6 hour period. During the following 16 hour period, the temperature shall be gradually reduced to 80 + 20°F. This constitutes one cycle. The relative humidity throughout the cycle shall be 95 + 5 percent. The cycle shall be repeated 10 times. Distilled or demineralized water having a pH value of between 6.5 and 7.5 shall be used. The air velocity through the test chamber shall not exceed 150 feet per minute. Following this test, the test set will be tested as specified in 4.4.3 through 4.4.6 and 4.4.8.

4.4.11.7 Pressure seal, case. The case with cover latched down shall be tested for watertightness as defined in MIL-STD-108, and method A shall be used to obtain a 3 foot water head.

4.4.11.8 Pressure seal, panel. The case with its cover off shall be evacuated to 70 + 3 mm/Hg below atmospheric pressure. The vacuum shall be drawn through one of the ten mounting screw holes with a special fixture having a good seal to the panel of the test set. After sealing off the test set and instrument, the vacuum shall be measured with a suitable vacuum measuring device. The vacuum shall not drop more than 2 mm/Hg in 15 minutes.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The test set shall be packaged in accordance with dwg 8822072 using the specified moisture barrier.

5.1.2 Level C. The test set shall be packaged in accordance with dwg 8822072 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The test set shall be packed in accordance with dwg 8822072.

5.3 Marking. Marking shall be as specified in accordance with dwg 8822072.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

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6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1683 Amendment 1 dated 19 September 1961.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1190-A060

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